

fareBeep | INTERNAL SYSTEM BLUEPRINT

1. THE SHARED LEDGER ALGORITHM

Every search initiated via 360dialog follows a Cache-First protocol. We check the Supabase 'fare_ledger' table before hitting external APIs. Data fresh for < 15 minutes is served immediately. This architecture reduces search latency by 80% and ensures unit economics stay positive as the user base scales.

2. THE TRANSACTIONAL HANDSHAKE (TIMER)

To mitigate 'Price Jumps', every booking is a timed session. A 10-minute expiry (expires_at) is written to the database. If the Paystack webhook confirms payment after the 10-minute window, the system triggers an automatic refund. A NGN 1,000 volatility buffer is maintained to absorb minor airline price changes during the payment window.

3. ENTERPRISE STACK (THE PLUMBING)

- Messaging: 360dialog (Meta Official BSP)
- Intelligence: Gemini 1.5 Flash (NLU Parser)
- Persistence: Supabase (PostgreSQL)
- Fare Inventory: Tiqwa (Local African GDS)
- Status: Aviationstack (ADS-B Live Tracking)
- Settlement: Paystack (HMAC-Verified Gateway)

4. FLIGHT STATUS LOGIC (RETENTION HOOK)

We initiate 'Status Watch' 3 hours before departure. Using Aviationstack data, we track the physical aircraft coming from the previous destination. If the aircraft is delayed, we send a 'Status Beep' to the user on WhatsApp instantly.